Results appendix

BTC/USDT 4h bars, 2018-01-01 to 2026-06-15. Walk-forward from 2022-01-01 with 19 quarterly refits; purge + 6-bar embargo. Test year (untouched during design): from 2025-06-15. Costs: 0.26% per round trip. Chosen model by the pre-registered rule: logreg. Edge validated on dev (kill switch): YES.

1. Does the model predict the trade outcome? (out-of-sample)

Model	Period	AUC target	AUC stop	Log-loss	Brier	IC exp. R
Base rate	dev	0.493	0.466	0.9469	0.5636	-0.052
Base rate	test	0.480	0.516	0.9542	0.5741	-0.001
Logistic	dev	0.612	0.572	0.9268	0.5498	0.030
Logistic	test	0.513	0.602	0.9476	0.5676	0.071
GBM (depth 3)	dev	0.581	0.554	0.9356	0.5551	0.002
GBM (depth 3)	test	0.545	0.530	0.9562	0.5747	0.001

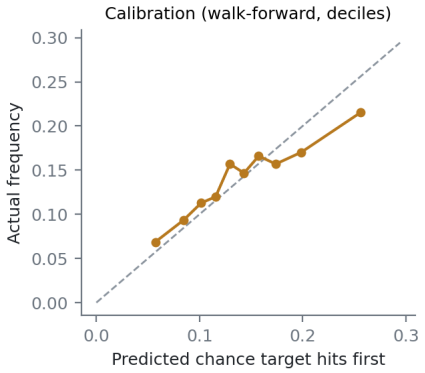
Outcome mix in the test year: target first 12.9%, stop first 30.8%, timed out 56.3%. IC = Spearman rank correlation between the model's expected result and the realised net result per bar.

2. Trading results, net of costs (one position at a time, 100% of account, no leverage)

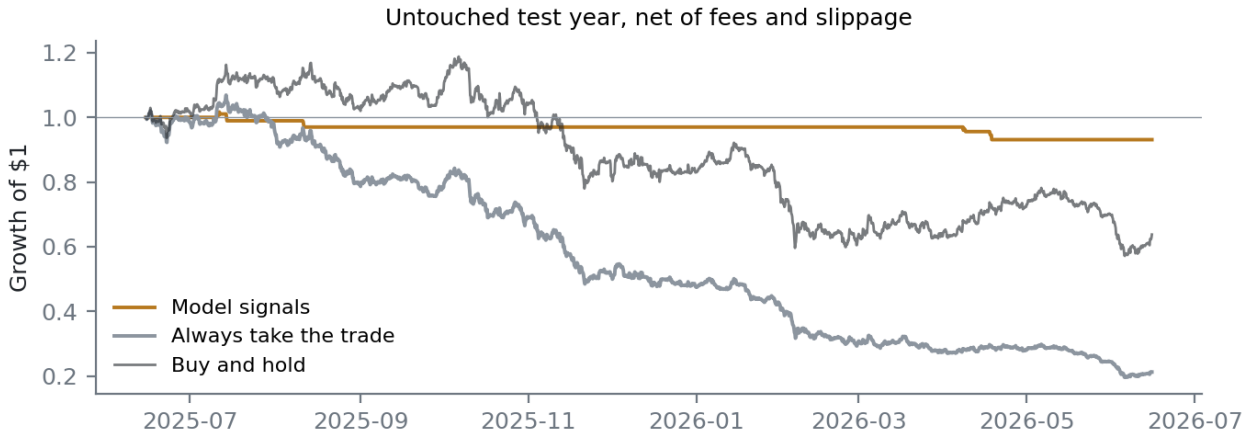
Strategy	Period	Trades	Win	Stop	Avg net R	CAGR	Sharpe	Max DD	Exposure
Model: all signals	dev	182	20.9%	21.4%	0.10	11.6%	0.71	-19.7%	11.8%
Model: all signals	test	5	0.0%	60.0%	-0.68	-6.9%	-1.83	-7.9%	1.0%
Model: Medium+ only	dev	78	21.8%	25.6%	0.05	0.9%	0.13	-15.5%	4.8%
Model: Medium+ only	test	0	–	–	–	0.0%	0.00	0.0%	0.0%
Always take trade	dev	1551	16.3%	29.1%	-0.10	-69.9%	-2.10	-98.5%	99.6%
Always take trade	test	454	13.2%	33.0%	-0.16	-78.8%	-3.79	-81.2%	99.9%
Buy and hold	dev	–	–	–	–	27.0%	0.71	-66.9%	100.0%
Buy and hold	test	–	–	–	–	-36.3%	-0.83	-51.2%	100.0%

Configurations tried: 8. Deflated Sharpe probability (dev): 0.575 (values near 1 mean the Sharpe is unlikely to be luck from trying several variants).

Costs	Model trades	Model CAGR	Model Sharpe	Always CAGR	Always Sharpe
0.0x	5	-5.7%	-1.65	-30.8%	-0.77
1.0x	5	-6.9%	-1.83	-78.8%	-3.79
2.0x	5	-8.1%	-1.96	-93.5%	-6.69









3. Honest track record by confidence tier (test year)

Tier	Signal bars	Target first	Stop first	Predicted R	Realised net R
Low	10	0.0%	40.0%	0.07	-0.47
Medium	0	–	–	–	–
High	0	–	–	–	–

Share of test-year bars with no signal: 99.4%.

4. Robustness

Check	Period	AUC target	IC	Trades	Sharpe	Always Sharpe
Barriers ×0.75	dev	0.572	0.023	96	-0.18	-2.72
Barriers ×0.75	test	0.508	0.078	1	-1.00	-4.25
Barriers ×1.25	dev	0.641	0.025	212	0.20	-1.80
Barriers ×1.25	test	0.511	0.074	13	-0.85	-3.17
Drop 'px_vs_ema'	dev	0.612	0.018			
Drop 'px_vs_ema'	test	0.505	0.091			

Year	AUC target	IC exp. R	Signal coverage	Signal avg net R
2022	0.626	-0.066	0.2%	0.80
2023	0.598	0.023	4.2%	0.05
2024	0.597	0.014	19.3%	0.05
2025	0.558	0.010	4.2%	-0.04
2026	0.550	0.141	0.7%	-0.76
Regime: calm	0.585	0.062		
Regime: normal	0.567	0.013		

Volatility model (QLIKE, lower is better): HAR 0.678 vs EWMA 0.735 on dev; HAR 0.539 vs EWMA 0.553 on test.

Largest logistic weights (target-vs-stop, standardised): px_vs_ema +0.911, ema_slope -0.354, park_24 -0.318, ret_6 -0.242, atr_pct +0.163, ret_24 -0.118.

5. Is it luck? (extra checks; none of these change the strategy)

Check	Dev	Test year
Model vs random entries (same trade count): p-value on Sharpe	0.000 (182 trades)	0.926 (5 trades)
Median Sharpe of random entries	-0.93	-0.32

Rank-IC (gross) and circular-shift p-value	0.030 (p=0.126)	0.071 (p=0.022)

Year	Trades	Target	Stop	Time-out	Net return
2022	3	2	0	1	12.1%
2023	32	8	7	17	7.0%
2024	117	22	24	71	31.7%
2025	33	6	10	17	-10.4%
2026	2	0	1	1	-4.0%

Same-candle ambiguity: 7 of 18,396 labels (0.04%) had one 4h candle touch both stop and target (counted as a stop); resolving them with 1h candles would flip 2.

5b. Release gate for the short-term buy signal (stricter, adopted from an independent review; dev period only)

Check	Result
At least 100 trades (182)	pass
Average net R +0.102, weekly block-bootstrap 95% lower bound -0.019 > 0	FAIL
Average net R at double costs -0.004 > 0	FAIL
Sharpe 0.71 better than taking every trade (-2.10)	pass

Overall: FAILED, so the short-term check runs in research mode and never says buy.

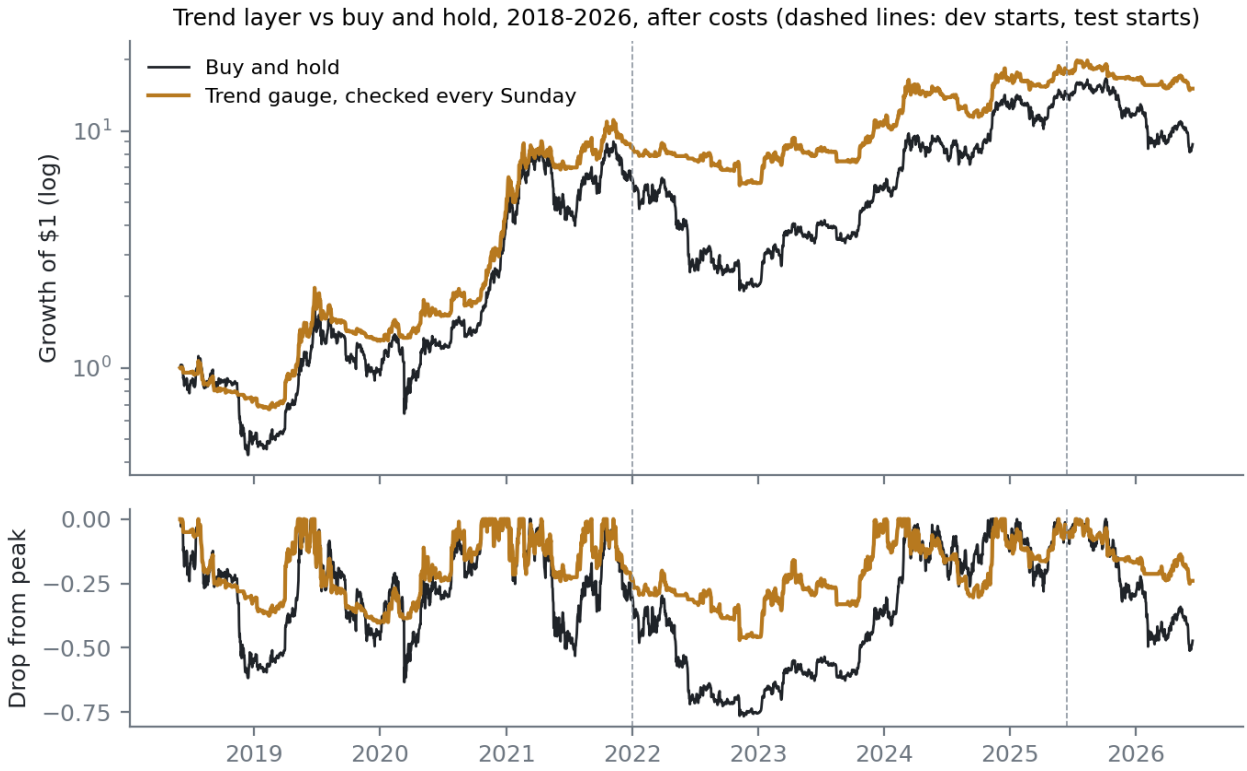
Short-term strategy with 1%-risk sizing	Trades	Total	Sharpe	Worst drop	Avg position
dev	182	17.2%	0.80	-6.9%	36.3%
test	5	-3.0%	-1.75	-3.5%	43.2%

7. The trend layer (time-series momentum, five lookbacks averaged, no tuning)

Variant	Period	Total	CAGR	Sharpe	Worst drop	Avg weight
Buy and hold	early	517.5%	66.2%	1.07	-63.4%	1.00
Buy and hold	dev	128.1%	27.0%	0.71	-66.9%	1.00
Buy and hold	test	-37.6%	-37.7%	-0.89	-51.2%	1.00
Buy and hold	full	778.4%	31.1%	0.75	-76.6%	1.00
Trend, Sundays only	early	748.9%	81.6%	1.47	-40.3%	0.57
Trend, Sundays only	dev	109.1%	23.8%	0.83	-33.8%	0.53
Trend, Sundays only	test	-15.0%	-15.1%	-0.75	-25.1%	0.43
Trend, Sundays only	full	1408.1%	40.2%	1.06	-47.0%	0.53
Trend, daily	early	669.0%	76.7%	1.42	-38.4%	0.57
Trend, daily	dev	115.9%	25.0%	0.88	-34.3%	0.54
Trend, daily	test	-10.6%	-10.7%	-0.50	-21.3%	0.44
Trend, daily	full	1383.5%	39.9%	1.06	-48.2%	0.54
Trend, vol-scaled	early	443.7%	60.4%	1.48	-33.2%	0.44
Trend, vol-scaled	dev	114.6%	24.8%	0.91	-31.8%	0.50
Trend, vol-scaled	test	-9.7%	-9.7%	-0.46	-20.5%	0.44
Trend, vol-scaled	full	953.5%	34.0%	1.09	-44.3%	0.47

Single lookback only (robustness)	Early Sharpe	Dev Sharpe	Test Sharpe	Full Sharpe	Full worst drop
7d	1.03	0.14	-0.65	0.57	-69.2%
14d	1.34	0.82	0.07	1.02	-52.8%
28d	1.44	0.84	-0.30	1.06	-56.1%
56d	1.48	0.99	-0.77	1.11	-44.5%
112d	1.07	0.94	-0.54	0.88	-60.0%

Sharpe difference vs buy and hold (Sundays-only trend), block bootstrap 95% interval: early +0.41 [-0.21, +1.01]; dev +0.11 [-0.49, +0.73]; test +0.14 [-1.00, +1.30]; full +0.31 [-0.09, +0.70]. Deflated Sharpe (full period, all trend variants counted): 0.989. Position changes per year with the Sunday check: 37. Buy and hold in this section is measured open to open on daily candles, so it differs slightly from section 2, which uses 4h candles (test year -37.6% vs -36.2%).



6. Data integrity

Source: data/btc_4h_data_2018_to_2025.csv. Raw rows 18512, clean rows 18510, duplicate-timestamp rows 4 (corrupted duplicates kept only if consistent with neighbours), missing 4h bars 16 (exchange outages, all before March 2020), labels dropped because their 24h window crossed a gap: 48. Every 4h candle was cross-checked against the independent 1h file: all but one corrupted 1h row agree within 0.2%.